The Islamia University of Bahawalpur
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The Islamia University of Bahawalpur

Final Year Project Proposal

Project Title: Smart Hotel Reservation and Management System

Nature of Project: ☑ Development ☐ Research

Area of Specialization:

☑ Artificial Intelligence (AI) ☐ Game Development ☐ Internet of Things (IoT)

☐ Machine Learning ☐ Mobile App Development ☐ Blockchain

☐ Data Science and Analytics ☑ Web Development ☐ Cybersecurity

☐ Image Processing ☐ Desktop Application

☐ Natural Language Processing (NLP) ☐ Other (Specify) _____

Introduction:

The hospitality sector depends on reliable room information, timely reservations, clear customer communication, and effective day-to-day hotel operations. Customers often need to compare accommodation according to destination, budget, travel dates, guest capacity, and facilities. Hotel operators, particularly smaller properties, may manage property details, room inventory, reservations, and financial records through separate or partly manual processes.

This project proposes a responsive, web-based platform that combines a multi-hotel marketplace with hotel management tools and AI-powered discovery. Participating hotels across Pakistan will be able to register, subject to administrator approval, and manage their own property profiles, room categories, prices, facilities, photographs, and

availability. Customers will search for suitable accommodation and submit reservations through a workflow that validates current availability and pricing on the server.

The system will provide two AI assistants: a global hotel finder that helps customers discover matching properties, and a hotel-specific assistant that answers questions about a selected property's authorized information. Structured database retrieval and, where appropriate, Retrieval-Augmented Generation (RAG) will ground AI responses in current hotel records. Hotel managers will also receive dashboards for reservations, occupancy, revenue, expenses, and estimated profit. The project will use Next.js, React, Tailwind CSS, and Appwrite, with server-side application logic and integration with a suitable large language model (LLM).

Problem Statement:

Customers may find it time-consuming to identify rooms that satisfy several constraints at once, such as destination, price range, dates, guest capacity, and required facilities. Information can be spread across different listings or channels, making direct comparison and confirmation of availability difficult. Hotel operators may also need to switch between separate processes to update rooms, handle reservations, and review business performance.

The project focuses on the following limitations commonly associated with fragmented or conventional workflows:

- Hotel discovery and comparison may require customers to search multiple sources or contact properties individually.
- Prices and room availability shown in a listing may need to be confirmed again before a reservation can be accepted.
- Hotel managers may lack a single dashboard combining room inventory, reservations, occupancy, revenue, and expenses.
- Generic AI answers can be unreliable if they are not grounded in current hotel data and may misstate prices, facilities, or policies.
- A shared platform must prevent one hotel's manager from viewing or modifying another hotel's private operational records.

Therefore, the proposed system will integrate hotel discovery, property-specific management, server-validated booking, database-grounded AI assistance, and reporting within a role-based multi-hotel application. The proposal identifies a design opportunity; it does not claim that every existing commercial platform has all of the limitations listed above.

Project Goal / Objectives:

The main goal of the Smart Hotel Reservation and Management System is to develop a secure, intelligent, and user-friendly web-based platform that simplifies hotel searching, room reservations, and hotel management across Pakistan. The system aims to connect customers with multiple registered hotels through a centralized platform where users can search, compare, and book rooms based on their budget, location, availability, and preferences. It will integrate AI-powered hotel recommendations and hotel-specific chatbots to provide accurate information about room availability, pricing, facilities, and booking policies. The system will also allow hotel owners to register their properties with administrator approval and manage rooms, reservations, customers, and hotel operations through individual dashboards. Additionally, it aims to automate booking and availability management, prevent double bookings, generate financial and operational reports, and ensure secure role-based access. By utilizing Next.js, React, Tailwind CSS, Appwrite, and AI technologies, the project seeks to improve customer experience, reduce manual hotel management tasks, and provide a scalable and efficient digital solution for the hospitality industry.

Application Architecture / Technology Stack / Tools / Hardware:

The system will use a client–server architecture. The Next.js application will provide the user interface and server-side API logic. Appwrite will provide authentication, database services, file storage, and permission controls. Server-side operations will validate user roles, hotel ownership, availability, and prices before changing protected records. The AI service will receive only the minimum authorized hotel context needed to answer the current question.

1. **Frontend**:Next.js, React, Tailwind CSS, HTML5, CSS3, JavaScript/TypeScript
2. **Backend services and database** : Appwrite Authentication, Database, Storage, and access permissions
3. **Server / application logic** : Next.js server-side routes and server actions/API endpoints, as appropriate
4. **AI:** A suitable LLM API; structured database retrieval for prices, dates, and availability; RAG where useful for descriptive information
5. **Charts and reports:** Recharts or an equivalent charting library**.**
6. **Tools:** Visual Studio Code, Git, GitHub, browser developer tools
7. **Testing:** Unit and integration test tools, manual functional testing, authorization test cases, and AI-response evaluation examples

8. **Hardware:** Laptop/desktop with a modern multi-core processor, 8 GB RAM minimum (16 GB recommended), stable internet, and a current web browser

Scope, Feasibility and Evaluation

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The Smart Hotel Reservation and Management System will provide a centralized, AI-powered web platform for customers, hotel owners, and administrators across Pakistan. The project scope includes hotel registration and administrator approval, hotel searching and filtering, room availability checking, online reservations, AI-powered hotel recommendations, and hotel-specific chatbots. Each hotel owner will have a separate dashboard to manage rooms, customers, reservations, facilities, revenue, expenses, and operational reports. The system will also implement secure authentication, role-based access control, and data isolation between hotels.

The project is technically and economically feasible because it will be developed using modern technologies such as Next.js, React, Tailwind CSS, Appwrite, and AI language models. These technologies provide suitable development tools, authentication, database management, cloud storage, and AI integration capabilities. The system can initially be developed and tested using free or low-cost development resources, making it suitable for a Final Year Project. Its modular architecture will support future improvements and expansion.

The system will be evaluated through functional testing, usability testing, performance testing, and security testing. Evaluation will focus on the accuracy of hotel recommendations, correctness of booking and room availability, prevention of duplicate reservations, chatbot response relevance, dashboard functionality, and protection of hotel-specific information. User feedback and test results will help determine the system's reliability, efficiency, and overall effectiveness in improving hotel reservation and management processes.

Student Data:

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8.1 Evaluation Criteria

- Functional testing: verify registration, approval, search, room management, reservation creation, cancellation/status updates, and reports.
- Booking correctness: test unavailable dates, changed prices, overlapping requests, and invalid guest/date inputs.
- Security testing: verify that customers cannot access other customers' reservations and hotel managers cannot access another hotel's records.
- AI grounding: test whether recommendations and answers match authorized database records; record unsupported or incorrect claims.
- Usability: conduct task-based walkthroughs for customer, hotel manager, and administrator roles and record issues for improvement.

9. Expected Outcomes and Deliverables

- A responsive multi-hotel reservation and management web application.
- Customer authentication, search, hotel listing pages, and reservation workflow.
- Hotel registration with administrator approval and role-specific dashboards.
- Property, room, price, facility, photograph, and availability management.
- Global AI hotel finder and hotel-specific AI assistant grounded in authorized data.
- Reports for reservations, occupancy, revenue, expenses, and estimated profit.
- Appwrite data model and access-control configuration.
- Architecture and database documentation, test cases, evaluation results, final report, and project demonstration.

10. Novelty and Significance

The project does not claim that online hotel booking, management dashboards, or conversational AI are individually new. Its intended contribution is the integration of a multi-hotel marketplace, separate property workspaces, budget- and date-aware discovery, two complementary AI assistants, backend-validated reservations, and integrated operational reports in a single prototype. It will also evaluate the grounding of AI responses and the enforcement of hotel-level data isolation.

References:

1. Lewis, P., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. Advances in Neural Information Processing Systems (NeurIPS). https://arxiv.org/abs/2005.11401
2. Vercel. Next.js Documentation. https://nextjs.org/docs
3. Appwrite. Appwrite Documentation. https://appwrite.io/docs
4. Meta. React Documentation. https://react.dev/
5. Tailwind Labs. Tailwind CSS Documentation. https://tailwindcss.com/docs

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